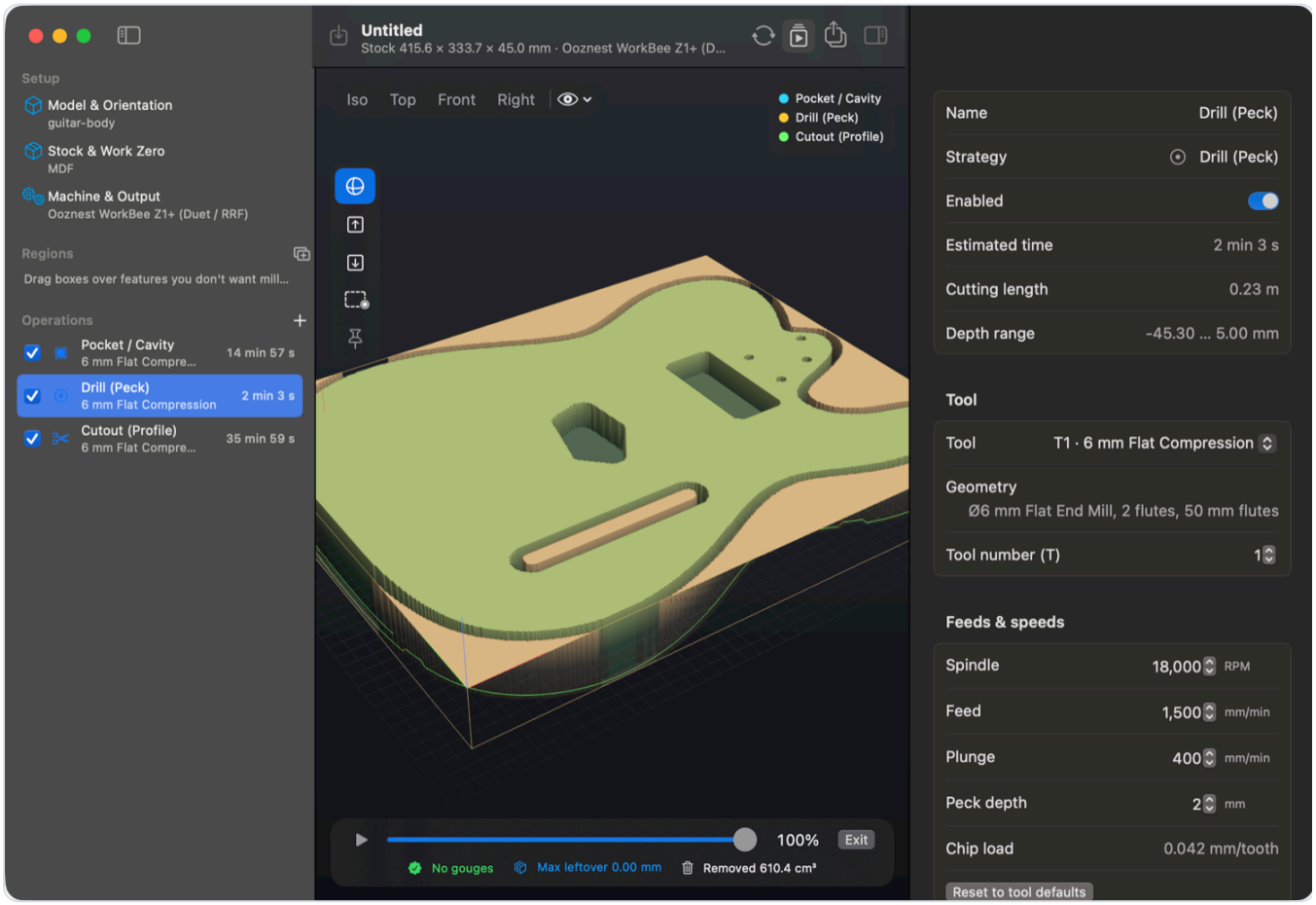
Swarf

User Guide · Mac-native CAM for hobby CNC



Free and open source under the MIT License · github.com/mouldiwarp/swarf

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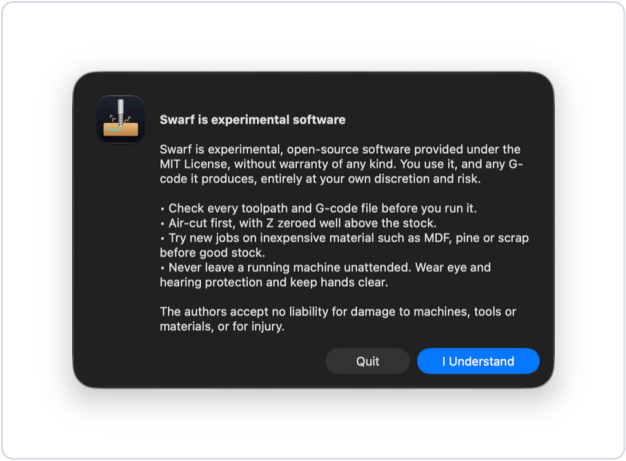
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1 · Before you start

> **Swarf is experimental software. You use it, and any G-code it produces, at your own discretion and risk.**
>
> It is provided under the MIT License without warranty of any kind. The authors accept no liability for damage to machines, tools or materials, or for injury.

Every time Swarf starts, it shows this notice and asks you to click **I Understand**. Clicking **Quit** closes the app. Make these habits part of every job:

- **Check every toolpath** in the 3D view and the simulator before you export.
- **Air-cut first**, with Z zeroed well above the stock, and watch the whole job.
- **Test new jobs on inexpensive material** such as MDF, pine or scrap before you cut good stock.
- **Measure your bits** and correct the tool library. Wrong diameters or flute lengths give wrong toolpaths.
- **Never leave a running machine unattended.** Wear eye and hearing protection and keep your hands clear.



The notice shown at every launch.

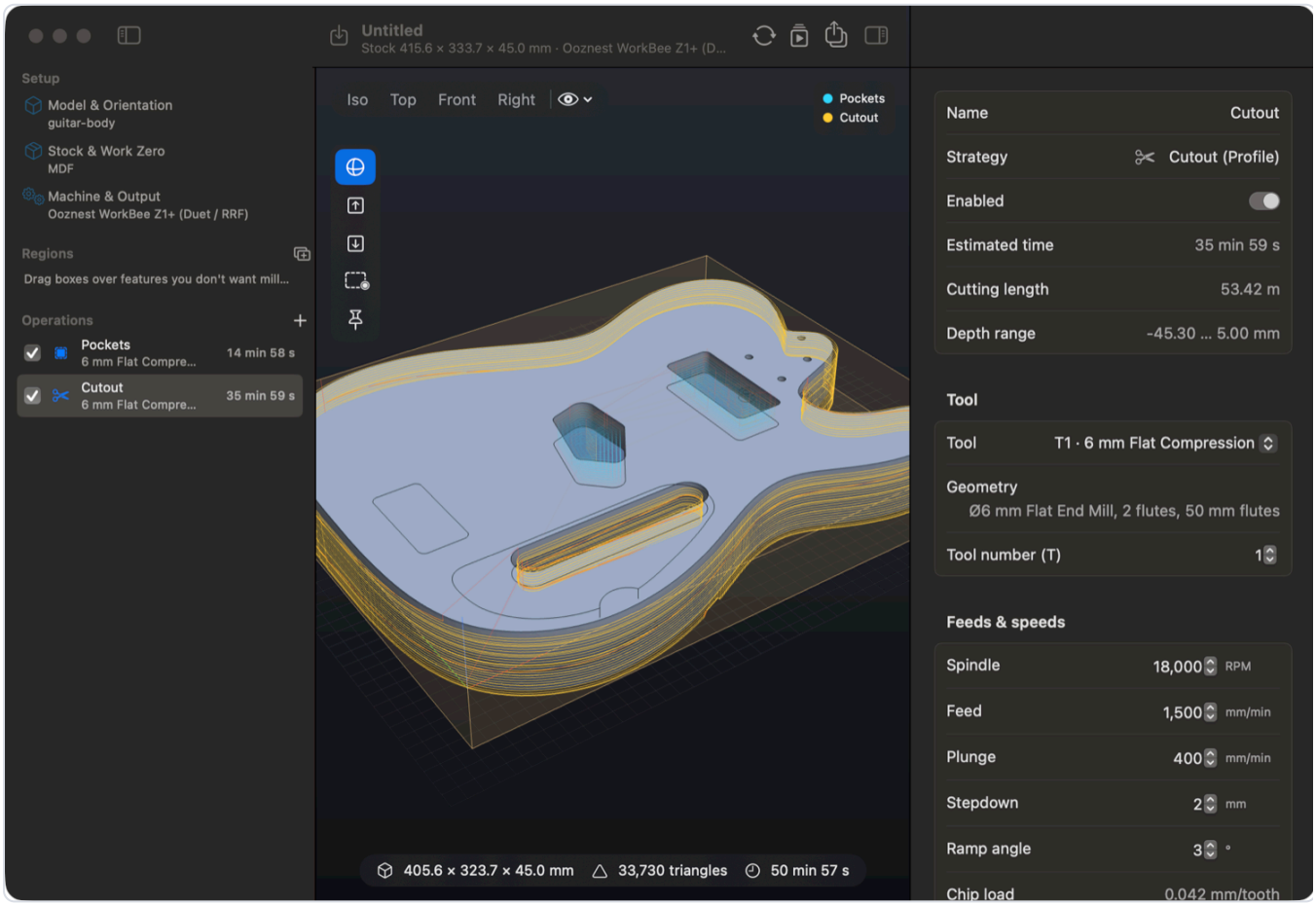
2 · Installing Swarf

Swarf runs on **macOS 15 or later** and builds from source with Xcode 16 or later. Install Xcode (or its command line tools), then in Terminal:

```
git clone https://github.com/mouldiwarp/swarf.git
cd swarf
scripts/build-app.sh          # builds build/Swarf.app
open build/Swarf.app
```

You can copy `build/Swarf.app` to your Applications folder. To update later, run `git pull` and `scripts/build-app.sh` again.

3 · The Swarf window



A guitar body with pocket and cutout toolpaths.

Area	What it does
Sidebar (left)	**Setup** (Model & Orientation, Stock & Work Zero, Machine & Output), **Regions** (keep-out boxes) and the **Operations** list. Tick or untick an operation to turn it on or off. Right-click to duplicate or delete.
Viewport (centre)	The 3D view of the model, the stock and the toolpaths. Use **Iso / Top / Front / Right** for preset views and the eye menu to show or hide layers. The tool strip down the left side switches between orbit, pick face up, pick face down, draw keep-out and place tabs.
Inspector (right)	The settings for whatever is selected in the sidebar.
Toolbar	Import STL, Generate (recalculate the toolpaths), Simulate, Export G-code, and show or hide the inspector.
Status bar	Model size, triangle count, a mesh health warning, and the total estimated run time.

Navigating the 3D view: drag to orbit, scroll or pinch to zoom, and drag with ⇧ or ⌥ held (or right-drag) to pan.

4 · Importing and orienting a model

1. Choose **File ▸ Import STL…** (⇧⌘I), click **Import STL** in the toolbar, or drop an STL on the window. Binary and ASCII STL both work. Swarf never changes your STL file.
2. STL files carry no units. Models smaller than 12 units are assumed to be in inches and scaled ×25.4 automatically. Under **Model & Orientation** you can set the scale yourself or use the **inch → ×25.4** button.
3. Turn the face you want to machine towards the cutter. Use **Lay Flat** (⌥⌘L) to put the largest flat face down, **Pick Face to Point Up** (⌥⌘U) or **Pick Face to Lay on Bed** (⌥⌘D) and click a face, or rotate in 90° steps about X, Y or Z (⌥⌘X / ⌥⌘Y / ⌥⌘Z).
4. Turn on **Highlight undercuts** to see faces a 3-axis machine can't reach from above. If important detail is highlighted, try a different side.

> A **Not watertight** warning in the status bar is usually fine. Swarf works from a height map, so small holes in the mesh don't matter. Check the 3D view for missing faces all the same.

5 · Stock, material and work zero

Select **Stock & Work Zero** in the sidebar.

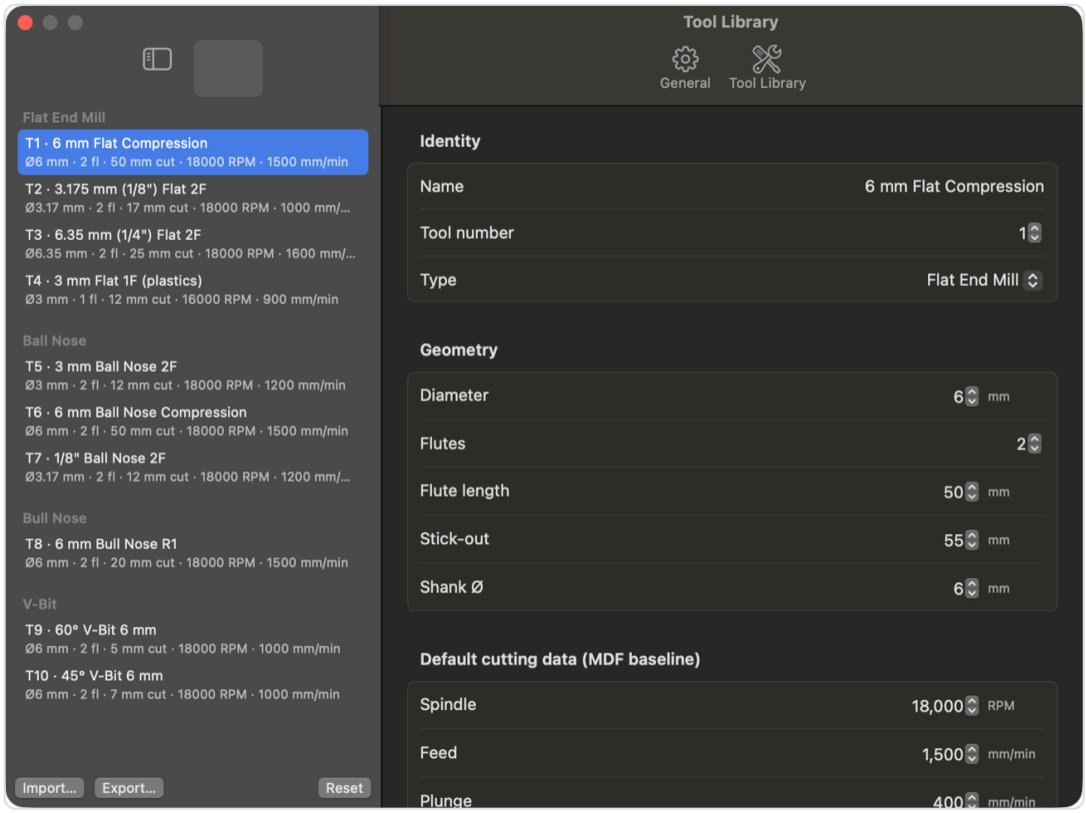
- **Material:** MDF, plywood, softwood, hardwood, acrylic, HDPE, foam or aluminium. The material scales each tool's default speeds and feeds (for example hardwood: feed and plunge ×0.8, stepdown ×0.75). Changing it later leaves existing operations as they are, and a banner offers to rescale them all.
- **Size from model + margins** sizes the stock automatically: set the XY margin and any extra on top or below. Turn it off to enter your real blank: Width (X), Depth (Y) and Thickness (Z).
- **Vertical** places the model at the top, centre or bottom of the stock. **Offset X / Y** shifts it within the blank.
- **Work zero**: pick one of 9 points on the stock, and set **Z zero** to the stock top or the bed. Swarf's default is the **centre of the left edge, top of stock**. Toolpaths start from the work zero, so the preview, the simulation and the G-code all begin at the point you choose.
- **Safe height** is how far above the stock the cutter travels between cuts.
- **Calculation resolution** (draft, normal or fine) trades accuracy for speed. Use draft while you experiment and normal or fine for the real job.

6 · Choosing your machine

Select **Machine & Output** to pick the post-processor. The default is the **Ooznest WorkBee Z1+ (Duet / RRF)**. Profiles are also included for generic RepRapFirmware, GRBL 1.1, Carbide 3D Shapeoko / Nomad, Sienci LongMill / AltMill, X-Carve, Genmitsu, Onefinity, grblHAL / FluidNC, MASSO G3, LinuxCNC, Mach3 / Mach4, UCCNC and Marlin. Each profile knows the machine's work envelope, maximum feeds and spindle range, and Swarf warns you if a job won't fit.

You can set the default machine, material and resolution for new projects in **Swarf ▸ Settings ▸ General**.

7 · The tool library



Swarf ▸ Settings ▸ Tool Library.

Click **Edit Tool Library…** in the Tool section of any operation's settings, or press ⇧⌘T (Window ▸ Tool Library…). It is also under **Swarf ▸ Settings…** (⌘,) ▸ **Tool Library**. Select a tool to edit it, then click **Save** (⌘S) to keep the changes or **Revert** to undo them. Until you save, the tool is marked "(edited)" and nothing else changes; if you pick another tool first, Swarf asks whether to save. **Add Tool** is at the bottom of the list, and Duplicate and Delete are on each tool's right-click menu. **Import…** reads Fusion 360 tool libraries and Swarf JSON files, and **Export…** saves your library. **Reset…** restores the starter set after asking.

- **Type:** flat end mill, ball nose, bull nose (with corner radius), V-bit (with included angle and tip diameter) or drill.
- **Geometry:** diameter, flutes, flute length (the cutting length), stick-out and shank diameter. Swarf warns you when a cut is deeper than the flute length.
- **Default cutting data** (an MDF baseline): spindle speed, feed, plunge, stepdown, stepover and ramp angle. The chip-load readout and warnings help you choose sensible values.
- **Tool number:** the T number written to the G-code.

The starter library includes 6 mm flat and ball-nose compression cutters (50 mm flutes), 1/8", 1/4" and 3 mm cutters, a 6 mm bull nose, 60° and 45° 6 mm V-bits, and 2 mm, 1 mm and 0.9 mm flat bits on 1/8" and 4 mm shanks for rest machining. New cutouts, pockets and drilling use the 6 mm flat, 3D finishing uses the 6 mm ball, and V-carving uses the 60° V-bit.

> When you edit a tool in the library, open projects that use it pick up the new geometry. Each operation keeps its own feeds and T number, so changing the library never silently changes the feeds of an operation you've already tuned.

8 · Operations

An operation is one machining step with one tool. Click **Full Processing** in the sidebar to let Swarf look at the model and add what it needs: facing if the model sits below the stock top, roughing and finishing for curved tops, pockets for cavities, and a cutout with tabs. Or click **+** to add operations yourself. Operations run in list order. Drag them to reorder.

Swarf recalculates toolpaths automatically when you change a setting or replace the model. To force a fresh calculation of every operation, click **Recalculate All Toolpaths** under the operation list, **Recalculate All** in the toolbar, or press ⌘G.

Operation	Use it for
Facing	Flattening the stock top. Set the depth and choose zig-zag or one-way passes.
Pocket / Cavity	Flat-bottomed recesses such as pickup and control cavities. Offset passes with an exact pass on every flat floor.
3D Roughing	Removing bulk material from curved shapes layer by layer, leaving stock for finishing.
3D Finish – Parallel	Smooth surfaces with a ball nose, in raster passes at any angle. Optionally only on slopes below a set angle.
3D Finish – Waterline	Steep walls, using constant-height contours. Optionally only on slopes above a set angle.
V-Carve	Lettering and logos with a V-bit (see chapter 12).
Drill (Peck)	Round holes that match the tool size (see chapter 10).
Cutout (Profile)	Cutting the part free with tabs (see chapter 9).
Dowel Holes (Flip)	Registration holes for machining both sides (see chapter 11).

Settings every operation has

- **Tool** from the library, and its **Tool number (T)**.
- **Feeds & speeds:** spindle RPM, feed, plunge, stepdown (the depth of each pass), stepover (the sideways step, as a percentage of the diameter) and ramp angle. **Reset to tool defaults** reloads them from the library. Library values are for MDF; for other materials Swarf scales them and says so in a banner over the 3D view, with the exact numbers.
- **Depth:** turn on **Limit maximum depth** and enter how far below the stock top this operation may cut. It works for every strategy (pockets, roughing, finishing, cutouts, holes, V-carving, facing), and material below the limit is left. In test-cut mode the shallower of the two limits wins.
- **Area:** boundary (whole stock, model silhouette plus a margin, or inside the model only) and an optional custom bottom height. With *inside the model only*, a hole that goes right through the part (a pickup slot, a pot hole) is not a wall: near a wall, the cutter may sweep up to a cutter width over the hole's edge, at the level of the floor around it. That clears floor strips narrower than the cutter, such as between a slot and a cavity wall. The middle of the hole is left alone.
- **Stock to leave:** radial (XY) and axial (Z) allowances for a later finishing operation.
- **Direction:** climb or conventional milling.

Changes regenerate the toolpaths automatically. Each operation shows its estimated time, cutting length, depth range and any warnings.

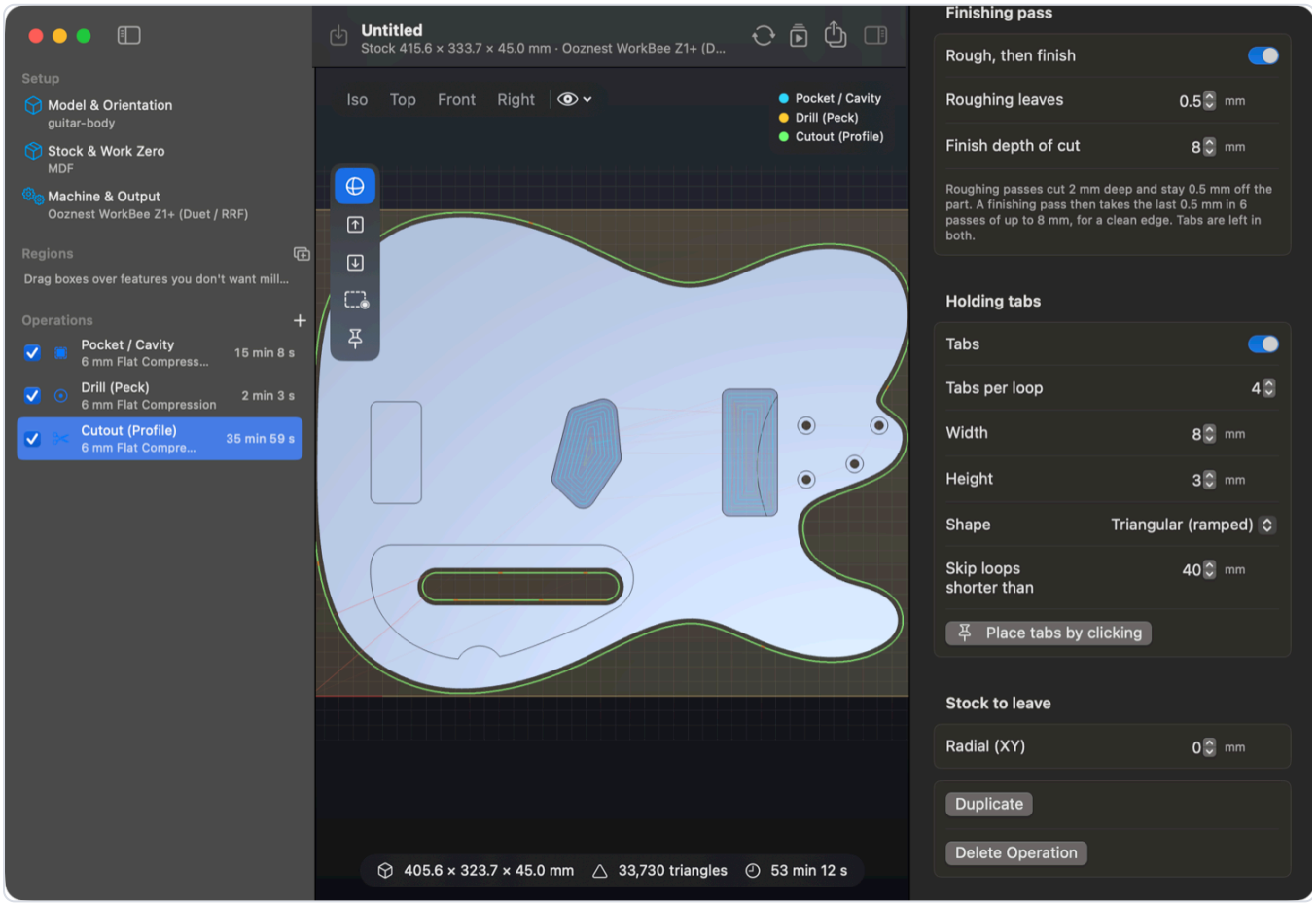
Rest machining: small bits for tight corners

A 6 mm bit leaves a 3 mm radius in every inside corner and can't enter a slot narrower than itself. Rest machining follows it with a smaller bit that cuts *only* what it left, so a rest pass takes minutes rather than re-cutting the whole pocket.

- Select the pocket or 3D roughing operation and choose **+ ▸ Rest Clearing**. Swarf adds a copy right after it with the largest flat bit in your library that is clearly smaller, set to follow the original tool. Change the bit in the copy's Tool section.
- Or, in any pocket, 3D roughing or 3D finishing operation, turn on **Only cut what a larger tool left** under **Rest machining** and pick the larger tool. A small ball nose can clean up concave curves after a larger one this way.
- The rest pass uses its own operation's area, depth and stock to leave, so keep them the same as the operation it follows.
- It works one corner or slot at a time, from the top down, and stays down between levels instead of lifting out each time.
- After a simulation, Swarf suggests a rest pass when a pocket or roughing operation leaves material that a smaller bit in your library could reach, with an **Add Rest Clearing** button.

> Small bits usually have a shank wider than the cutter: a 0.9 mm bit on a 1/8" or 4 mm shank. Swarf keeps that shank off the walls above the flutes, so beside a wall the bit only reaches its flute length below the top of the wall. Anything deeper is left, and the operation's warnings say how much. Enter your bits' real flute length and shank diameter in the tool library; the starter values are typical, not measured.

9 · Cutouts, finishing passes and tabs



Cutout settings with roughing passes, a finishing pass and holding tabs.

The cutout follows the outside of the model, offset by the tool radius, and goes through the stock by **Cut below stock** (0.3 mm by default). **Also cut internal holes** cuts through-openings first, while the part is still held at the edges.

Rough, then finish

On by default. The **roughing** passes cut at your normal stepdown but stay **0.5 mm** (**Roughing leaves**) off the part. A **finishing pass** then follows the true outline and removes that last 0.5 mm skin at a much deeper **Finish depth of cut**, 4× the roughing stepdown by default. With a 2 mm stepdown, the finishing pass takes 8 mm per pass. Because it only removes 0.5 mm sideways, the load stays light and the edge comes out clean. If you change the finish depth, a **4×** button puts it back to the default.

Wide trench for chip clearance

Off by default. In a deep cut the slot is only as wide as the cutter, so chips pack in, get cut again and heat the bit. Tick **Wide trench** (Chip clearance) and every roughing level is cut twice: first a pass further out from the part, then the normal pass next to it. The slot becomes **Trench width** × the tool diameter (1.5× by default: 9 mm for a 6 mm bit) and chips have room to clear. The tool stays down between the two passes, and the finishing pass is unchanged.

- **Second pass feed**: the pass next to the part only cuts part of the bit's width, so it runs faster than the outer pass, 1.25× the feed by default. Set it to 1 to keep one feed throughout.

- Roughing still takes much longer: on a guitar body with a 6 mm bit, 35 minutes without the trench, 58 with it (63 if the second pass isn't faster). Keep it for deep or chip-prone cuts.
- The stock needs that much more room around the part. Side B and dowel holes allow for the wider slot.
- Holes too small for the trench, and outlines too near the stock edge, get a plain slot; the operation's warnings say how many.
- Tabs cross the whole width of the trench.

Holding tabs

- **Tabs per loop** places tabs automatically and moves them onto straight sections where possible. Or click **Place tabs by clicking** and click on the outline.
- **Width**, **Height** (above the stock bottom) and **Shape** (triangular/ramped or rectangular).
- **Skip loops shorter than** stops small slugs from getting tabs.
- Both the roughing and the finishing passes leave the same tabs. Cut them with a flush-trim saw or chisel and sand them flush.

10 · Peck drilling

Drill (Peck) finds round vertical holes in the model and drills the ones that match the tool, within **Hole size match ±** (0.5 mm by default). Holes of other sizes are listed in a warning. Larger through holes are cut by the cutout, and larger blind holes need a pocket.

- **Peck depth** (the operation's stepdown, half the tool diameter by default) is how far each peck goes, at the plunge rate.
- **Full retract** backs out to 1 mm above the stock after every peck to clear chips, then drops quickly back to 0.5 mm above the last depth. Use it for deep holes.
- **Chip break** just lifts a little (**Chip-break lift**) to snap the chip. It's faster for shallow holes.
- Blind holes stop at their floor. Through holes go below the stock by **Through holes: below stock**. A pointed drill goes deeper still, so its full diameter comes out of the bottom face.
- The output is plain G0/G1 moves, so it runs on any firmware.

> Drill **before** the cutout, so the part is still held while it's drilled.

11 · Two-sided machining (flip)

Some parts need work on both faces, such as a guitar body with a carved top and control cavities in the back, or a neck. Swarf machines them in four steps. The stock is located by two dowels, so the X/Y zero never changes.

1. **Dowel holes.** With the stock clamped down, two 6 mm holes are pecked through the stock and 5 mm into the spoilboard. They come in pairs mirrored across the stock's centre line (front to back), so each hole lands on a dowel again after the flip.
2. **Side A (the top).** The top is machined and the part cut out with tabs, as in a one-sided job.
3. **Flip.** Lift the stock, push 6 mm dowels into the spoilboard holes, roll the stock over front-to-back (about the X axis; the ends stay where they are) and drop it onto the dowels. Clamp it. Keep the X/Y zero and re-probe Z on the new top.
4. **Side B (the back).** The back is machined: cavities, carving and holes.

Setting it up

- Turn on **Machine both sides (flip)** under **Stock & Work Zero**. A **Dowel holes** operation is added at the top of the list.
- Select the **Dowel holes** operation to set where the holes go (see *Placing the dowel holes* below).
- In the Dowel holes inspector, **Into spoilboard** (5 mm by default) and **Clearance from part** set the hole depth and how far the holes stay from the cutout. The holes are as deep as the stock plus the spoilboard depth. Swarf refuses holes deeper than the cutter's flute length, so the collet can't reach the stock; with a 50 mm cutter, 45 mm stock is about the limit.
- Set an operation's **Side** to **B** in its inspector, or right-click it and choose **Move to Side B**. Pockets, 3D roughing and finishing, and drilling can run on side B. Cutouts, facing and V-carving can't: the part is already cut out from side A.
- **Full Processing** on a two-sided job adds the dowel holes, side A as usual (finishing with the cutout), and side B: pockets for flat-floored cavities, or roughing and finishing for a carved back.
- The sidebar lists the operations under **Dowel holes**, **Side A** and **Side B**, in the order they run.

Placing the dowel holes

By default the placement is **Automatic**. Swarf puts one hole on the centre line in the waste beyond each end of the part. If the part fills the stock length, it puts a mirrored pair where the gap between the part and the stock edge is widest, such as either side of a guitar body's waist. If neither fits, it says there is no room: increase the **XY margin** or the stock size, or place the holes yourself.

To place them yourself, select the Dowel holes operation and click **Place by Clicking** (or the ◎ button in the viewport tool palette). The view switches to Top. Hover over the stock to preview a hole and its mirror on the other side of the centre line:

- **Green**: both holes are clear of the part (by the cutout slot plus the clearance) and of the stock edge. Click to add the pair.
- **Red**: one of the holes lands on or too close to the part or the edge. Clicking shows why and adds nothing.

- Click an existing hole to remove its pair. **Use Automatic** clears your holes. Press **Done** or Esc to stop placing.

The inspector lists every hole with its X/Y position (from the stock's front-left corner). Two holes, one pair, are enough; add a second pair for more support on a large part. If a later change (a bigger cutter, a new orientation) moves the part onto a hole, the hole turns red and the operation reports an error until you move it.

What side B leaves alone

The part hangs in the stock by the tabs from side A's cutout. Side B never cuts the frame of stock around the part (it holds the dowels and the clamps), and never cuts where the tabs are. Keep-out boxes are drawn on the side-A view and apply to side B in the same place on the part, over their full depth.

Viewing and simulating

Side-B toolpaths are drawn under the part, where they cut. Use **Bottom** view (⌘4) to look at the back as it lies after the flip. The simulation cuts side B from below and checks the back for gouges too.

Exporting

Each step is written to its own numbered set of files, such as `Body_1_dowels.gcode`, `Body_2_sideA.gcode` and `Body_3_sideB.gcode`. Each file starts with comments saying what to do before running it. Side-B coordinates are what the machine sees after the flip.

12 · V-carving

V-Carve engraves with a V-bit. The bit follows the centre line of each shape and goes deeper where the shape is wider, so edges and corners stay sharp. **Engrave** chooses what is carved:

- **Recesses**: areas sunk into the model's top face, such as text in a plaque.
- **Silhouette**: the model's whole outline, such as an STL of extruded lettering, carved into the stock top.

Limit depth (flat bottom) caps the depth for wide shapes. Stepdown splits deep carves into several passes. Use keep-out or machine-only regions to limit what gets engraved. A 60° bit carves deeper and narrower than a 90° bit, and a 45° bit deeper still. Set the bit's tip diameter in the library if it isn't sharp.

13 · Keep-out regions

Keep-out boxes protect features you don't want milled, or cover clamps.

1. Click the draw-keep-out tool in the viewport strip (or the button next to **Regions**, ⌥⌘K) and drag a box in the 3D view.
2. Click a box to select it. It turns orange. Drag it to move it, nudge it with the arrow keys (1 mm; ⇧ 10 mm; ⌥ 0.1 mm), or press ⌫ to delete it. Click empty space to deselect.
3. In the inspector, set exact coordinates, switch the type to **Machine only within** (which limits operations to the inside of the box), or turn on **Allow cutting down to a height** to protect only what's below a set Z.

The cutter stays a full radius away from keep-outs in every operation except cutouts. The simulator reports any keep-out violations.

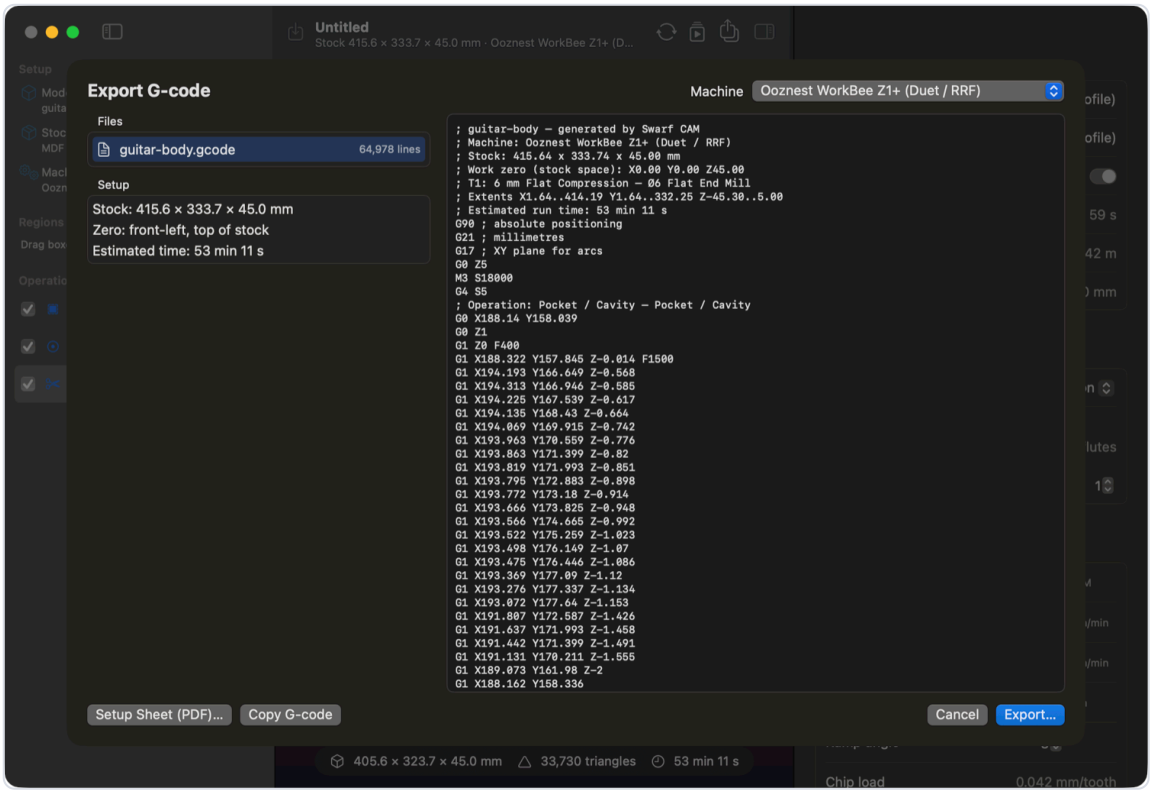
14 · Simulation and checks

Click **Simulate** (⌘R) to remove material from a virtual block of stock. Use the scrubber or play button to watch the job. Below the scrubber you'll see:

- **Gouges**: places cut deeper than the model, shown in red. A good job says **No gouges**.
- **Max leftover**: material left behind, shown as a blue heat map. Expect some in inside corners smaller than the tool.
- **Removed** volume and any **keep-out violations**.

Click **Exit** to return to the toolpath view. Warnings such as feeds out of range, cuts deeper than the flute length, a job outside the machine envelope, or cuts below the stock bottom appear on each operation in the inspector.

15 · Exporting G-code



The export sheet, with a preview of the WorkBee G-code.

1. Click **Export G-code** (⌘E). Check the machine, the list of files, the stock size, the zero point and the run time.
2. Read the preview. The header lists the stock, the work zero, each tool and the run time.
3. Type a name in **File name** if you don't want the model's name. The file list shows the result, and the project remembers the name for next time. Click **Export…** and choose a folder. **Setup Sheet (PDF)…** saves a printable sheet with the stock, the zero point, a top view and the tool list. **Copy G-code** puts the program on the clipboard. Tick **Show G-code** to see the first 400 lines of the selected file; it is off by default, and Swarf remembers your choice.

For the WorkBee, Swarf writes **one file per tool**, numbered in the order they run. Other machines get a single file with tool-change commands where the controller supports them.

16 · Running the job on a WorkBee

1. Fix the stock down, keeping clamps outside the toolpaths (or covered by keep-outs).
2. Fit the first tool. Jog to the **work zero shown in the setup sheet** (by default the centre of the stock's left edge) and set X and Y zero.
3. Probe or touch off Z on the **top of the stock**.
4. Upload the first file in the Duet web interface. For the first run, raise Z zero 10–20 mm and air-cut the start of the job.
5. Run the file. Stay at the machine with your hand near the emergency stop.

6. For each further file: change the bit, re-probe Z (keep X and Y), and run it.

17 · Troubleshooting

Problem	What to try
"Operation produced no toolpath" or "Nothing to clear"	The tool may be too big for the feature, or the boundary excludes it. Try a smaller tool, or change the Area boundary.
Drilling skips holes	The holes don't match the tool diameter. Raise **Hole size match ±**, choose a matching tool, or use a pocket.
"Cut depth exceeds the tool's flute length"	The job is deeper than the bit can cut. Use a longer bit, or check the flute length in the tool library.
"Cuts below the stock bottom"	Check the stock thickness and model placement. For V-carving, set a depth limit or use a wider-angle bit.
Chip-load warnings	Low chip load rubs and burns, so raise the feed or lower the RPM. High chip load stresses the bit, so do the opposite.
Generating is slow	Use draft resolution while setting up and switch to normal or fine for the final job.
Model too small or large	It was probably modelled in inches or metres. Set the scale under Model & Orientation.

18 · Command line

`swarf-cli` runs the same engine without the app:

```
swift build -c release
.build/release/swarf-cli model.stl --out ./gcode --simulate --png preview.png
.build/release/swarf-cli model.stl --ops pocket,drill,cutout --material plywood
.build/release/swarf-cli --list-machines
```

Options: `--machine`, `--material`, `--quality draft|normal|fine`, `--lay-flat`, `--rotate x:90`, `--ops` (facing, pocket, roughing, parallelFinish, waterlineFinish, vcarve, drill, cutout), `--keepout x0,y0,x1,y1`, `--vcarve-source recesses|silhouette`, `--vcarve-depth`, `--margin`, `--flip [mm]` (two-sided job; prefix side-B operations with `b:`, e.g. `--ops dowelHoles,pocket,cutout,b:pocket`), `--rest <mm>` (a rest pass with the flat bit closest to that size after each pocket and 3D roughing), `--trench <x>` (cutouts rough a slot *x* × the tool diameter wide), `--trench-feed <x>` (feed of its second pass, default 1.25×).

19 · Keyboard shortcuts

Action	Shortcut	Action	Shortcut
Import STL	⇧⌘I	Lay flat	⌥⌘L
Recalculate all toolpaths	⌘G	Pick face to point up	⌥⌘U
Simulate	⌘R	Pick face to lay on bed	⌥⌘D
Export G-code	⌘E	Rotate 90° about X / Y / Z	⌥⌘X / ⌥⌘Y / ⌥⌘Z
Settings / tool library	⌘,	Draw keep-out	⌥⌘K
Undo / Redo	⌘Z / ⇧⌘Z	Move / delete selected keep-out	Arrows / ⌫

Iso / Top / Front / Right view	⌘0 / ⌘1 / ⌘2 / ⌘3	Bottom view (side B)	⌘4

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